

Unified Healthcare Management System

ERD Requirements

1. Hospitals

- The healthcare network has a number of hospitals located in several governorates.
- **Attributes:** Hospital_ID, Hospital_Name, Hospital_Type, Location.

2. Departments

- Each hospital contains one or more departments.
- **Attributes:** Department_ID, Department_Name.
- A department must belong to one and only one hospital.
- Each department always has one doctor assigned to manage it.
- A doctor cannot manage more than one department at the same time.

3. Doctors

- Each department is assigned one or more doctors.
- **Attributes:** Doctor_ID, Doctor_Name, Specialty, Phone.
- A doctor must be assigned to one and only one department.

4. Patients

- The system stores information about all patients.
- **Attributes:** Patient_ID, Patient_Name, Gender, Birth_Date, Phone.

5. Visits

- A patient can have one or more visits.
- **Attributes:** Visit_ID, Visit_Date, Visit_Type, Symptoms, Total_Cost.
- Each visit must belong to one and only one patient.
- Each visit must be handled by one doctor.
- Each visit must take place in one hospital.

6. Medical Records

- The system stores a medical record for each visit.
- **Attributes:** Record_ID, Diagnosis_Notes, Treatment_Notes, Follow_Up_Required.
- Each medical record belongs to one and only one visit.

7. Prescriptions

- Doctors can issue prescriptions during visits.
- **Attributes:** Prescription_ID, Prescription_Date, Notes.
- Each prescription belongs to one visit.
- Each prescription is issued by one doctor.

8. Drugs

- The system stores information about all drugs.
- **Attributes:** Drug_ID, Drug_Name, Drug_Category, Unit_Price.
- A prescription may contain one or more drugs.
- A drug may appear in many prescriptions.
- For each drug in a prescription, the system stores: Dosage, Frequency, and Quantity.

9. Drug Inventory

- Each hospital maintains a drug inventory.
- **Attributes:** Inventory_ID, Quantity_Available, Reorder_Level, Expiration_Date.
- Each inventory record belongs to one hospital and one drug.
- A hospital may store many drugs, and a drug may exist in many hospitals.

10. Insurance Claims

- The system processes insurance claims submitted by patients.
- **Attributes:** Claim_ID, Claim_Date, Claim_Amount, Approved_Amount, Claim_Status.
- Each claim belongs to one patient.
- Each claim is generated from one visit.
- A patient may have many claims.

11. Feedback

- Patients can submit feedback about hospitals and doctors.
- **Attributes:** Feedback_ID, Rating, Comments, Feedback_Date.
- Each feedback belongs to one patient.
- Each feedback is related to one hospital and one doctor.
- A hospital and a doctor may receive many feedback records.

12. Beds

- Each hospital contains zero or more beds.
- **Attributes:** Bed_ID, Bed_Number, Bed_Type, Status.
- Each bed belongs to one and only one department.
- A bed can be assigned to one patient at a time.
- A patient may be assigned to zero or one bed per visit.

14. Diagnoses

- The system stores diagnoses recorded during visits.
- **Attributes:** Diagnosis_ID, Diagnosis_Code, Diagnosis_Name, Diagnosis_Date, Notes.
- Each diagnosis belongs to one and only one visit.
- A visit may have one or more diagnoses.
- Each diagnosis is recorded by one doctor.